

Supplementary Figures: Strain–Doping Coupling in Graphullerene

Xingqiang Chen
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Companion to the main manuscript on coupled (ϵ, δ) response in qHP C₆₀. Figures follow the observables vocabulary of Capobianco *et al.* [1] (localization, λ ,

J , FCWD). **DATA:** Fig. S5 Marcus λ panels refresh from `exp9.polaron_verification.json` after vertical SP converge; until then literature λ schematics only.

[1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa,

Nano Letters **24**, 7891 (2024), key reference from refer-info.md - polaron transport mechanisms.

qHP C₆₀ graphullerene ($\epsilon = 0\%$, pristine)

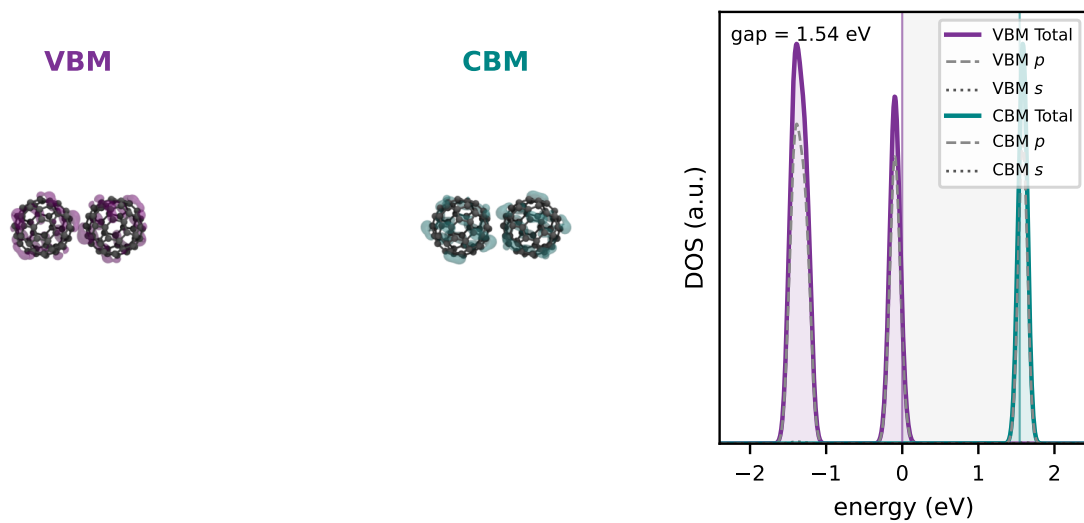


FIG. 1. VBM/CBM isosurfaces and π -resolved DOS (Exp. 7). [pending: VMD] when PDF absent.

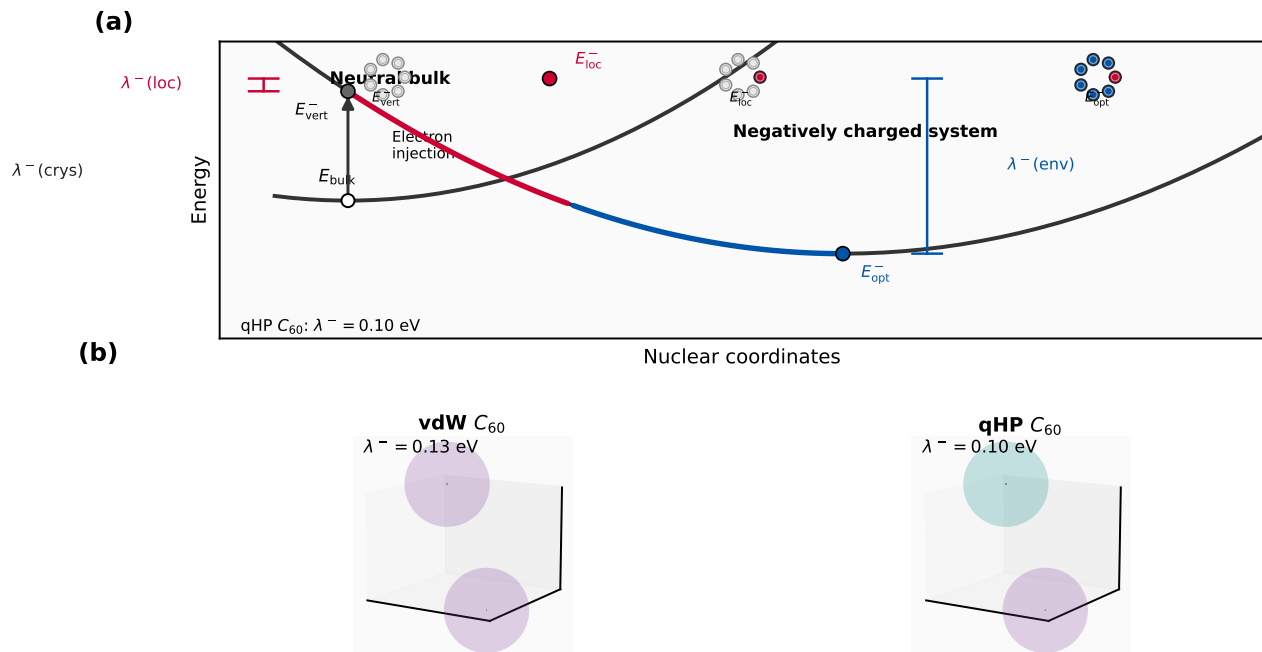


FIG. 2. Marcus reorganization schematic (Capobianco-style configuration coordinate and vdW vs. qHP comparison). Exp. 9 DFT λ from vertical – adiabatic charged relaxations: [pending: 5 GEO_OPT + 8 vertical SP]; see `derived.lambda_eV` in audit JSON.

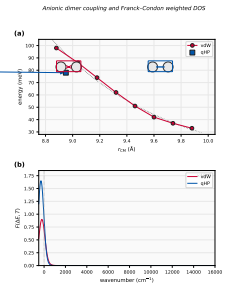


FIG. 3. Nearest-neighbor coupling J and Franck-Condon weighted density (FCWD) for vdW vs. qHP dimers. [synthetic **FCWD**] from literature mode frequencies; MolFC replacement planned. Exp. 4 supplies two verified J points at tetramer scale (PBE+D3).